PLTW POE

Activity 3.1.4 Equivalent Resistance

PLTW

EQUIVALENT RESISTANCE

Follow the Rules!

GOALS

RESOURCES

- Solve for equivalent resistance in series and parallel circuits.

GOALS

RESOURCES

- **Equivalent Resistance (Microsoft Excel)**
- **Equivalent Resistance (Google Sheets)**
- **Activity 3.1.4 Equivalent Resistance (Downloadable PDF)**

By now, you can successfully describe the mathematical rules that govern both current and voltage in series and parallel circuits. But what about resistance? Does it follow similar rules in each of these circuits?

In this activity, you'll model more circuits to determine these mathematical relationships.

Equivalent Resistance in Series

1. Determine the total resistance of the circuit in Figure 1. Use mathematics to justify your response.

   HINT: Use Ohm's Law.

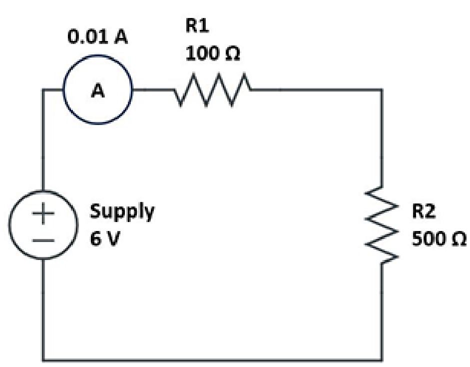

Figure 1. Series Circuit

The total resistance you calculated for Figure 1 is also called the **equivalent resistance**. You could replace the 100 Ω and 500 Ω resistors with one 600 Ω resistor, to make a circuit with properties equivalent to those of the original circuit. In other words, equivalent resistance is equal to the sum of the individual resistors in the circuit. Following is the mathematical rule that governs equivalent resistance in a series circuit:

$$R_{eq} = R_1 + R_2 + \dots + R_n$$

Time to Practice

2 Calculate the equivalent resistance of a series circuit with 330 Ω, 500 Ω, and 1500 Ω resistors.

3 Calculate the equivalent resistance of the circuit in Figure 2.

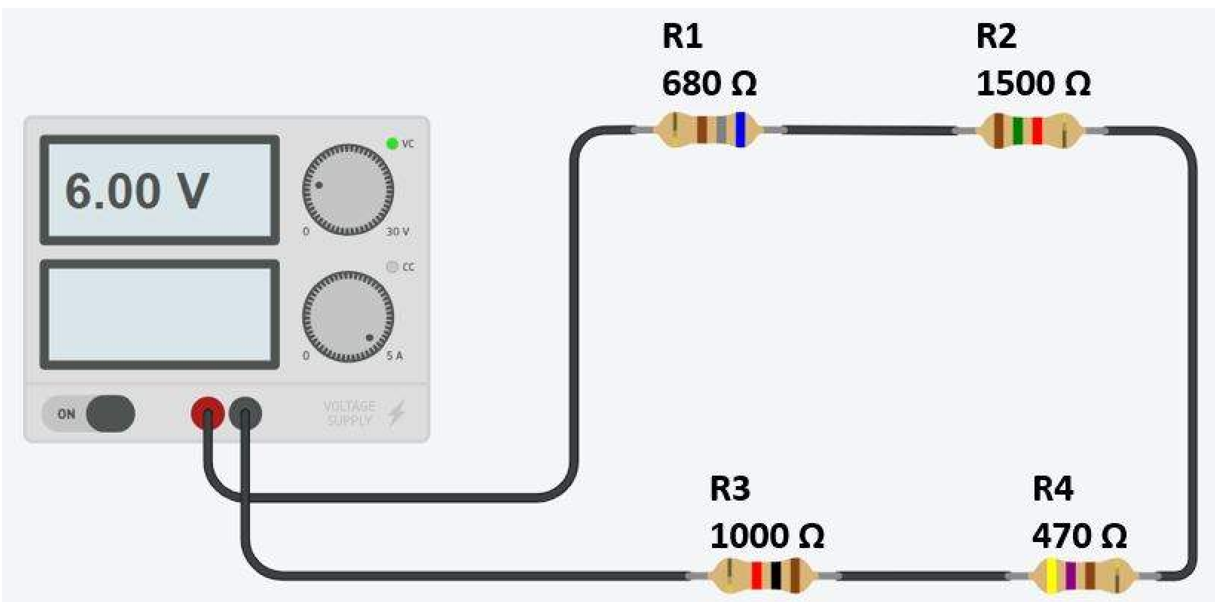

Figure 2. Practice Series Tinkercad® Circuit

4 Calculate the equivalent resistance of the circuit in Figure 3.

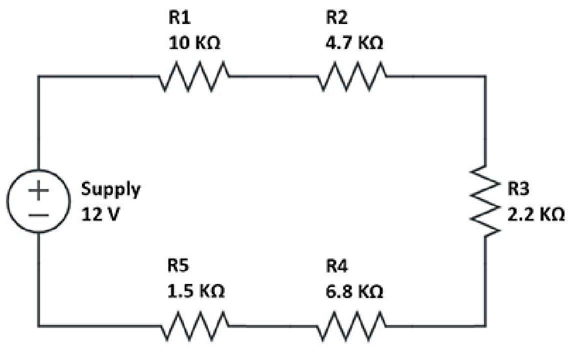

Figure 3. Practice Series Circuit Schematic

Equivalent Resistance in Parallel

The equivalent resistance in a series circuit is relatively straightforward. But does the same mathematical rule apply to parallel circuits?

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Determine the total resistance of the circuit in Figure 4. Use mathematics to justify your response.

HINT: Use Ohm's Law.

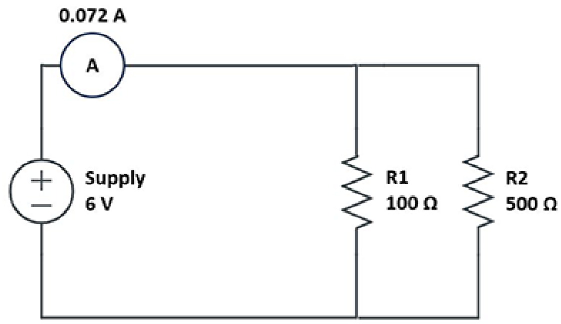


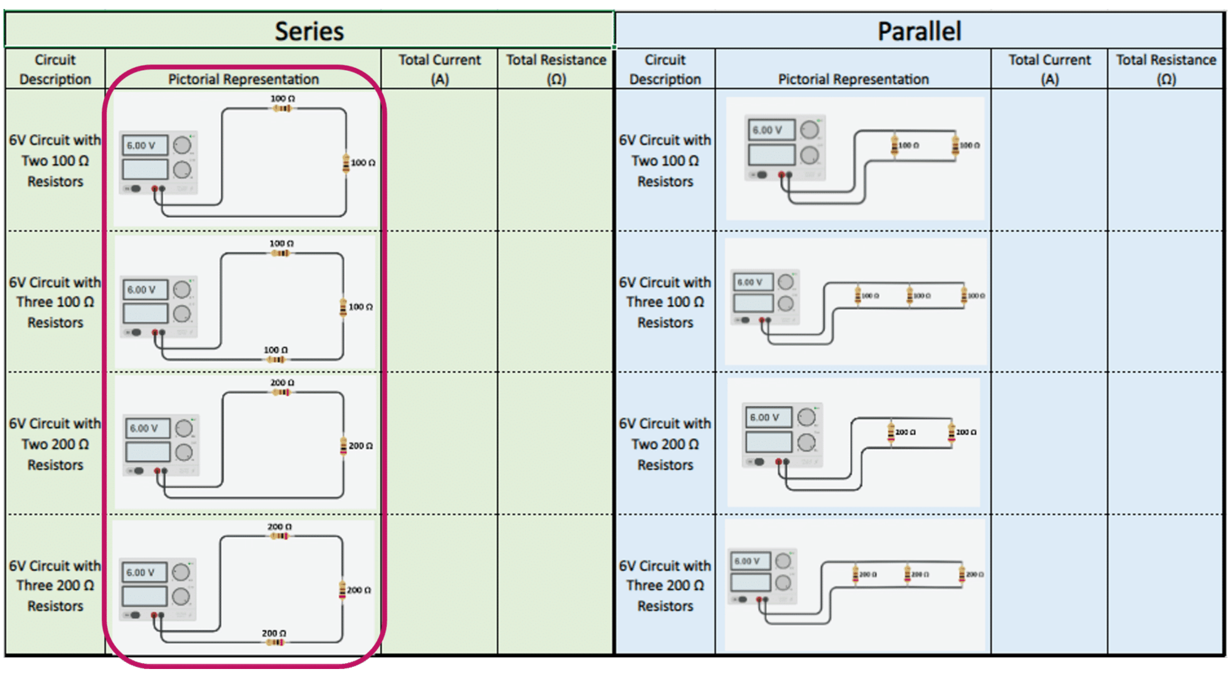
Figure 4. Parallel Circuit

As with the series circuit, the total resistance you calculated is also the equivalent resistance. You could replace the 100 Ω and 500 Ω resistors with one 83.3 Ω resistor to make a circuit with properties equivalent to those of the original circuit.

But how is it possible that the equivalent resistance is less than the resistance of each individual resistor in the circuit? Time to model some circuits and measure circuit parameters to develop a mathematical rule for the equivalent resistance in a parallel circuit.

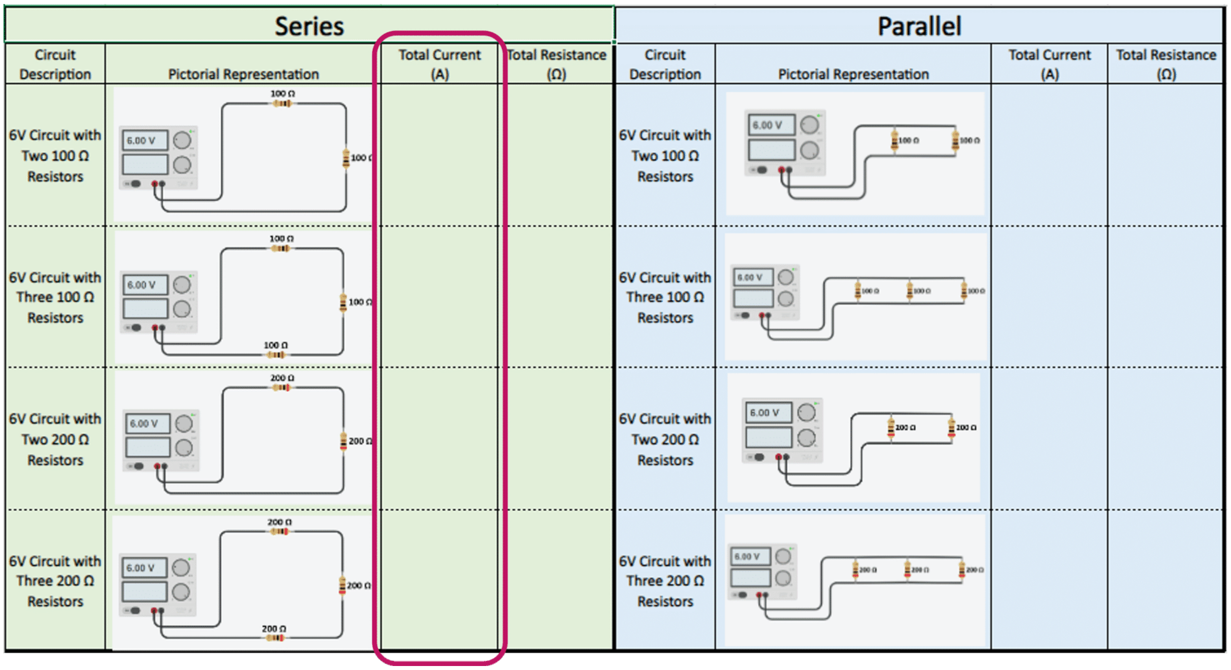
6. Download and save **Equivalent Resistance (Microsoft Excel)** or **Equivalent Resistance (Google Sheets)**.

7. Review the four series circuit descriptions on the spreadsheet in the Pictorial Representation column. Then, use Autodesk® Tinkercad® to model each of these series circuits.



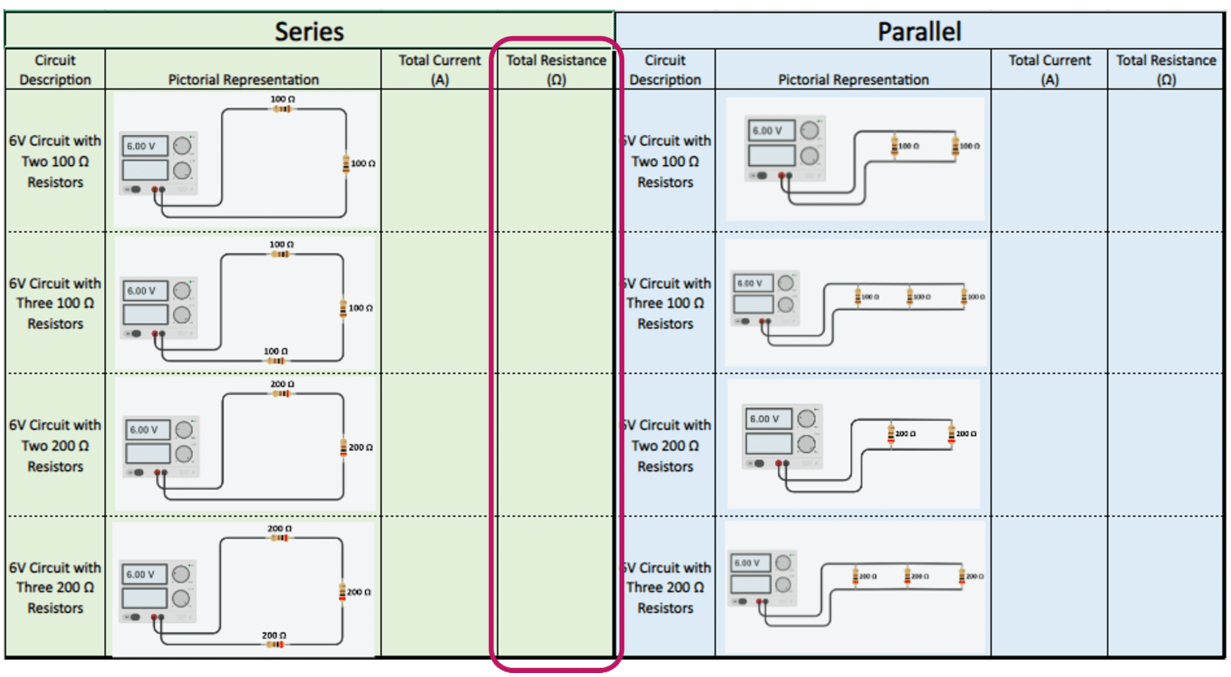
Series				Parallel			
Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)	Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)
6V Circuit with Two 100 Ω Resistors				6V Circuit with Two 100 Ω Resistors			
6V Circuit with Three 100 Ω Resistors				6V Circuit with Three 100 Ω Resistors			
6V Circuit with Two 200 Ω Resistors				6V Circuit with Two 200 Ω Resistors			
6V Circuit with Three 200 Ω Resistors				6V Circuit with Three 200 Ω Resistors			

8. For each circuit, use a multimeter to measure the total current in the circuit. Record these values in the Total Current column of the spreadsheet.



Series				Parallel			
Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)	Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)
6V Circuit with Two 100 Ω Resistors				6V Circuit with Two 100 Ω Resistors			
6V Circuit with Three 100 Ω Resistors				6V Circuit with Three 100 Ω Resistors			
6V Circuit with Two 200 Ω Resistors				6V Circuit with Two 200 Ω Resistors			
6V Circuit with Three 200 Ω Resistors				6V Circuit with Three 200 Ω Resistors			

9 Use Ohm's Law to calculate each circuit's total resistance. Record these values in the Total Resistance column of the spreadsheet.



Series				Parallel			
Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)	Circuit Description	Pictorial Representation	Total Current (A)	Total Resistance (Ω)
6V Circuit with Two 100 Ω Resistors				6V Circuit with Two 100 Ω Resistors			
6V Circuit with Three 100 Ω Resistors				6V Circuit with Three 100 Ω Resistors			
6V Circuit with Two 200 Ω Resistors				6V Circuit with Two 200 Ω Resistors			
6V Circuit with Three 200 Ω Resistors				6V Circuit with Three 200 Ω Resistors			

10 Repeat steps 7–9 for the four parallel circuits on the right side of the spreadsheet.

- Model each of the parallel circuits in Tinkercad®.
- Measure and record the total current for each circuit and record values in the Total Current (A) column.
- Use Ohm's Law to calculate each circuit's total resistance and record values in the Total Resistance (Ω) column.

Reflection

How is it possible to add resistors to a circuit, yet decrease the overall equivalent resistance?

Deriving the Equation

You can use Ohm's Law to derive the equation for equivalent resistance in a parallel circuit. Consider the circuit in Figure 5.

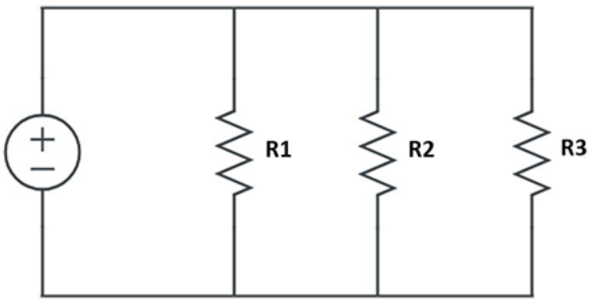


Figure 5. Parallel Circuit

11. Recall and record how to calculate the total current in a parallel circuit.

12. In your equation from step 11, substitute an expression for current in terms of voltage and resistance.

13. Divide each side of your equation by voltage (V).

NOTE: Remember, the voltage across each branch of a parallel circuit is equal to the total voltage.

14 Now solve for R_{eq}.

Time to Practice

15 Calculate the equivalent resistance of a parallel circuit with 330 Ω, 500 Ω, and 1500 Ω resistors.

16 Calculate the equivalent resistance of the circuit in Figure 6.

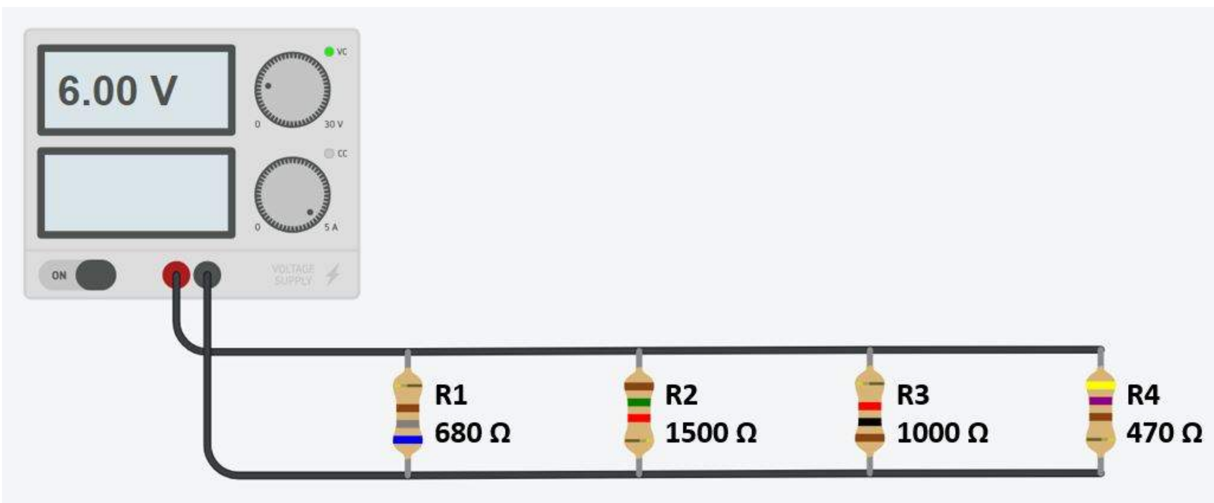


Figure 6. Practice Parallel Tinkercad® Circuit

17 Calculate the equivalent resistance of the circuit in Figure 7.

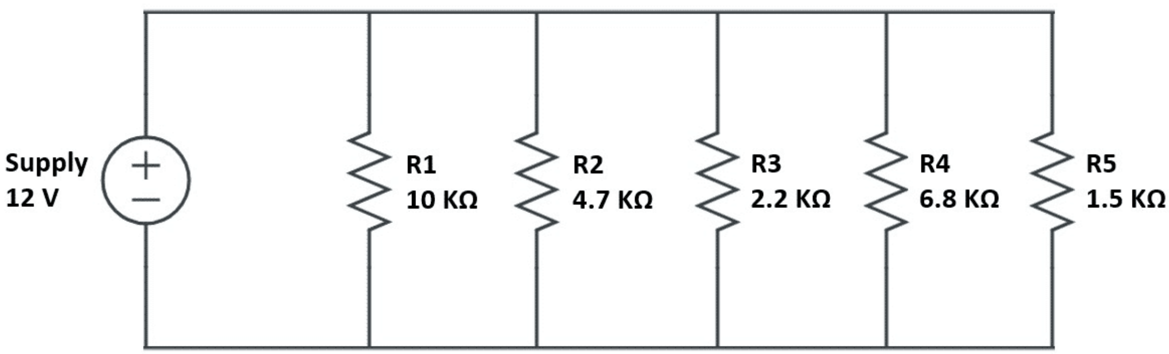


Figure 7. Practice Parallel Circuit Schematic

Combination Circuits

You've developed mathematical equations to calculate the equivalent resistance for series and parallel circuits. However, most circuits are not simply one or the other. They might contain elements of both series and parallel circuits. Consider the circuit in Figure 8.

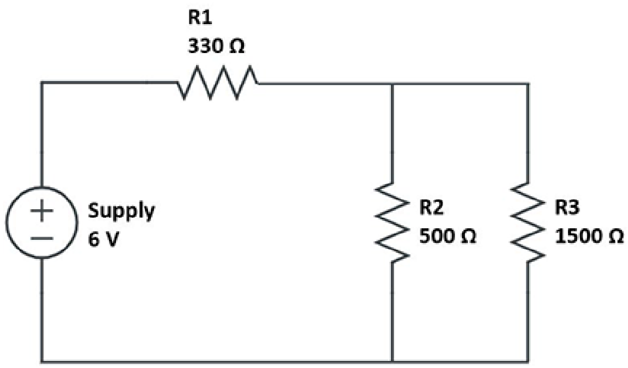


Figure 8. Combination Circuit

Notice that R1 is in series with the power supply. After R1, the circuit has two parallel branches, one with R2 and one with R3. Those branches come back together and return to the power supply. This type of circuit is known as a **combination circuit**.

To solve for the equivalent resistance of the circuit, simplify the circuit one step at a time until you are left with a series circuit.

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Calculate an equivalent resistance for R2 and R3 in Figure 9.

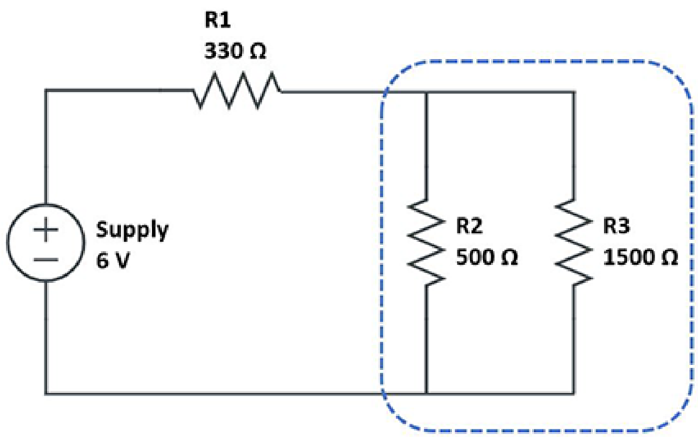

Figure 9. Parallel Branches

19 Redraw the circuit schematic, substituting an equivalent resistor for the two parallel branches.

20 Calculate the equivalent resistance of the circuit using the rule for series circuits.

NOTE: It is good practice to check your work. In this case, you can build the circuit in Tinkercad® and measure the total current. Using the total current, the voltage on the power supply, and Ohm's Law, you can calculate the equivalent resistance to check your work.

21 In Tinkercad, model the circuit in Figure 8. Be sure to accurately set the values of the resistors, and the voltage of the power supply.

22 Measure the total current of the circuit.

23 Use Ohm's Law to calculate the equivalent resistance of the circuit.

Conclusion

Question 1

Calculate the equivalent resistance of the circuit in Figure 10.

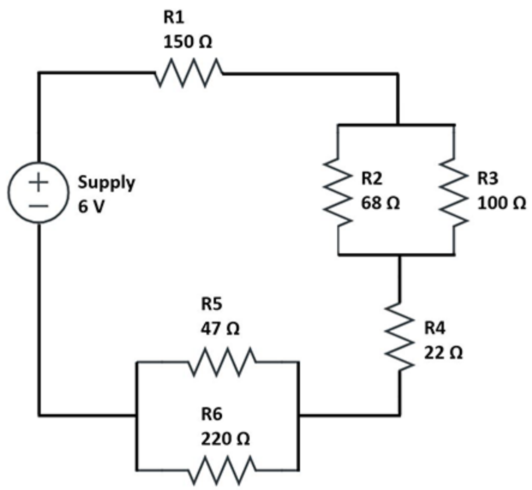

Figure 10. Combination Circuit

Question 2

Given the equivalent resistance you calculated for the circuit in Figure 10, calculate the total current for the circuit.